Tsung-Yen Tsai 蔡宗諺

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Physical AI and robotics systems engineer with 4+ years building simulation, planning, and optimization systems for industrial automation. Turns physical-world constraints into verifiable workflows, with current technical-lead work across 3D hookup routing and semiconductor layout planning.

EXPERIENCE

MetAI

Taipei, Taiwan | Apr 2025 - Present

Technical Lead, Hookup & Layout Engines

Jul 2026 - Present

Python, Taichi, CUDA, USD, Optuna, PyTorch, Isaac Sim, Omniverse

- Led productization of a 3D industrial routing foundation, designing immutable layered voxel-world and request-specific routing-mask artifacts with Taichi CUDA occupancy processing, deterministic graph search, and reproducible parameter optimization.
- Defined the product architecture and reusable-core / customer-workspace boundary for a semiconductor layout engine spanning domain modeling, deterministic rule verification, and placement workflows.

Senior AI Reinforcement Learning Engineer

Apr 2025 - Present

- Re-architected the RL training environment for semiconductor tool placement by separating layout data, rule execution, and result rendering, improving reproducibility and reducing layout inspection time by 93%, from about five minutes to 20 seconds.
- Built a 0-to-1 3D routing system that turns dense semiconductor layouts and constructability rules into multi-pipe routing candidates for customer and engineering review, treating spatial reasoning as search-and-verify rather than rendering.

TAC Dynamics

Taipei, Taiwan | Apr 2024 - Apr 2025

Senior Robotics Engineer

C#, Python, gRPC, Redis, PostgreSQL, DES

- Co-defined domain boundaries for an AS/RS warehouse automation platform, helping consolidate 150 microservices into 5 core modules and reduce system-wide memory usage by 30%.
- Redesigned and implemented the AS/RS fleet-management system as a modular monolith, replacing distributed coordination with in-process communication and shared state to simplify development and latency-sensitive robot control; validated the system through a month-long hardware integration test with real payload operations.
- Owned the core FMS integration for an AMR customer deployment, defining the MES interface and integrating the customer's production system with the AMR fleet through successful on-site deployment.
- Built a discrete-event simulation platform modeling order flow, inventory, container allocation, stack locking, reshuffle, and retrieval logic, enabling randomized comparison of warehouse strategies before full hardware and real-time simulation were available.

FARobot

Taipei, Taiwan | Feb 2022 - Mar 2024

Algorithm Engineer

C++17, Python, ROS 2, Unity

- Built a hybrid FMS-and-simulation validation platform supporting 100+ simulated AMRs and multi-day continuous reliability testing, so field changes could be reproduced and de-risked before deployment.
- Improved throughput 40%+ at a panel-manufacturing AMR site by reconstructing operations from field logs, testing changes in simulation, and validating releases on site with the deployment team.
- Developed and deployed a MAPF-based route planner for a tight multi-robot environment, separating planning from execution and adding localization margins for stable real-field operation.

Vanderlande · Robotics Engineer Intern

Veghel, Netherlands | Jul 2021 - Sep 2021

EDUCATION

Eindhoven University of Technology

Eindhoven, Netherlands | 2019 - 2021

M.Sc. in Mechanical Engineering, Control Systems Technology

Thesis: Situational-Aware Motion Planning with a Shared Semantic World Model

National Taiwan University

Taipei, Taiwan | 2014 - 2018

B.S. in Mechanical Engineering